COLUMBIA UNIVERSITY
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STAT 5703

Statistical Inference and Modeling for Data Science

Dobrin Marchev

GLM

A Generalized Linear Model (GLM) extends the classic linear regression model in two ways:

1. $Y|\boldsymbol{x}$ ~ Exponential family

2. A function between the expected outcome and the predictors:

$$g[\mathrm{E}(\mathrm{Y}|\mathbf{x})] = \boldsymbol{x}'\boldsymbol{\beta}$$

the function $g()$ is called the *link function.*

Exponential Family

A family of univariate pdf/pmf is said to be an exponential if it can be expressed as:

$$f(x;\theta) = c(\theta)h(x)e^{t(x)q(\theta)}$$

where $c(\theta) \geq 0$, $h(x) \geq 0$.

An alternative reparametrization is known as the *canonical* form:

$$f(x;\eta) = h(x)e^{\eta t(x) - \psi(\eta)}$$

In the above, η is known as the *natural* parameter.

Definition

In either of the forms:

$$f(x;\theta) = c(\theta)h(x)e^{t(x)q(\theta)}$$

or

$$f(x;\eta) = h(x)e^{\eta t(x)-\psi(\eta)}$$

$t(X)$ is called the natural *sufficient statistic*.

- Sufficiency attempts to formalize the notion of no loss of information.
- A sufficient statistic is supposed to contain by itself all the information about the unknown parameters of the underlying distribution that the entire sample could have provided.
- In that sense, there is nothing to lose by restricting attention to just a sufficient statistic in one's inference process.

Example

Bernoulli:

$$f(x;p) = p^x(1-p)^{1-x} = (1-p)\left(\frac{p}{1-p}\right)^x$$
$$= (1-p)e^{x\log\frac{p}{1-p}}$$

so, to match the form $f(x;p) = c(p)h(x)e^{t(x)q(p)}$, in this case:

$$c(p) = (1-p), h(x) = 1, t(x) = x, q(p) = \log\frac{p}{1-p}$$

For the canonical form,

$$f(x;p) = e^{x\log\frac{p}{1-p} - \log\frac{1}{1-p}} = e^{x\eta - \log(1+e^\eta)}$$

where $\eta = \log\frac{p}{1-p}, p = \frac{e^\eta}{1+e^\eta}, 1 + e^\eta = \frac{1}{1-p}$

Exercises

- Derive the sufficient statistic for a sample

$X_1, \ldots, X_n \sim$ Bernoulli(p)

- Verify the Poisson density is also of exponential type

Results

If $f(x;\eta) = h(x)e^{\eta t(x) - \psi(\eta)}$ is in canonical form, then:

a) $E\big(T(X)\big) = \psi'(\eta)$
b) $Var\big(T(X)\big) = \psi''(\eta)$
c) $M(t) = e^{\psi(\eta+t) - \psi(\eta)}$

Exercise: Verify the results for the Binomial distribution by comparing them to the well-known answer using the traditional parametrization.

Multivariate Definition

Let $X = (X_1, \ldots, X_n)$ be a random vector with joint pdf

$$f(x; \boldsymbol{\theta}) = c(\boldsymbol{\theta}) h(\boldsymbol{x}) e^{\sum_{j=1}^{k} t_j(\boldsymbol{x}) q_j(\boldsymbol{\theta})}$$

where $\boldsymbol{\theta}$ is a k-parameter vector.

Such distribution is said to be in a k-parameter exponential family.

Example

Let $X_1, \ldots, X_n$ be a random sample from N(μ, σ).
Then $\theta = (\mu, \sigma^2)$ and

$$f(\boldsymbol{x}; \mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$$

$$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{n\mu^2}{2\sigma^2}} e^{-\frac{1}{2\sigma^2}\left(\underbrace{\sum_{i=1}^{n} x_i^2}_{t_2(\boldsymbol{x})} - 2\mu \underbrace{\sum_{i=1}^{n} x_i}_{t_1(\boldsymbol{x})}\right)}$$

T

o match with $f(x; \boldsymbol{\theta}) = c(\boldsymbol{\theta}) h(\boldsymbol{x}) e^{\sum_{j=1}^{k} t_j(\boldsymbol{x}) q_j(\boldsymbol{\theta})}$, we have:

$$q_1(\mu, \sigma^2) = \frac{\mu}{\sigma^2}, q_2(\mu, \sigma^2) = -\frac{1}{2\sigma^2}$$

$$h(\boldsymbol{x}) = (2\pi)^{-\frac{n}{2}}, c(\mu, \sigma^2) = \sigma^{-n} e^{-\frac{n\mu^2}{2\sigma^2}}$$

Exercise

Let $X_1, \ldots, X_n$ be a random sample from $\Gamma(a, b)$.
Write the likelihood in such a form to show that it belongs to the two-parameter exponential family.

GLM

More specifically, GLM assumes that the response variable has a *generalized linear model* (aka exponential dispersion model EDM) pdf:

$$f(y_i;\theta,\phi) = a(y_i,\phi)e^{\frac{y_i\theta-\kappa(\theta)}{\phi}}, i = 1,\dots,n$$

where ϕ is called a *dispersion parameter* (and could be known).
$\kappa(\theta)$ is called cumulant function.
This form of the distribution is aka *natural* exponential family, because θ is the natural parameter.
Notation: Y ~ EDM(μ, ϕ), where E(Y_i) = μ

Note: For a *fixed* value of the dispersion parameter ϕ it is a one-parameter exponential family (indexed by θ).

Example: Normal distribution

The normal distribution with unknown mean μ and variance σ^2:

$$f(y;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} = \frac{e^{-\frac{y^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} e^{\frac{y\mu - \frac{\mu^2}{2}}{\sigma^2}}$$

Match with:

$$f(y;\theta,\phi) = a(y,\phi) e^{\frac{y\theta - \kappa(\theta)}{\phi}}$$

$\theta = \mu$ is the natural aka canonical parameter.

$\kappa(\theta) = \frac{\mu^2}{2} = \frac{\theta^2}{2}$ is the cumulant function

$\phi = \sigma^2$ is the dispersion parameter

$a(y,\phi) = \frac{e^{-\frac{y^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}$ is the normalizing function

Other examples are Exponential, Gamma, Binomial, …

GLM: Moment generating and cumulant functions

Theorem:

$$M(t) = e^{\frac{\kappa(\theta+t\phi)-\kappa(\theta)}{\phi}}$$

$$K(t) = \frac{\kappa(\theta + t\phi) - \kappa(\theta)}{\phi}$$

Example:

Normal distribution $\kappa(\mu) = \frac{\mu^2}{2}$

$$K(t) = \frac{(\mu + t\sigma^2)^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} = \mu t + \frac{\sigma^2 t^2}{2}$$

GLM: Mean and variance

Theorem:

$$E(Y) = \mu = \frac{d\kappa(\theta)}{d\theta}$$

$$Var(Y) = \phi \frac{d^2\kappa(\theta)}{d\theta^2}$$

where $\frac{d^2\kappa(\theta)}{d\theta^2} = \frac{d}{d\theta}\left(\frac{d\kappa(\theta)}{d\theta}\right) = \frac{d\mu}{\mathrm{d}\theta}$

We can define the variance function

$$V(\mu) = \frac{\mathrm{d}\mu}{\mathrm{d}\theta} \Rightarrow Var(Y) = \phi V(\mu)$$

Example: Normal distribution $\kappa(\theta) = \frac{\theta^2}{2}$

$$E(Y) = \frac{d\left(\frac{\theta^2}{2}\right)}{d\theta} = \theta = \mu, V(\mu) = 1 \Rightarrow Var(Y) = \sigma^2$$

GLM: Unit deviance

Suppose we want to write the EDM $f(y;\theta,\phi) = a(y,\phi)e^{\frac{y\theta-\kappa(\theta)}{\phi}}$ as a function of the mean μ. Denote

$$t(y,\mu) = y\theta - \kappa(\theta)$$

Then $t(y,\mu)$ has a unique max w.r.t. μ at $\mu = y$. This allows us to define a very important quantity, the unit deviance:

$$d(y,\mu) = 2[t(y,y) - t(y,\mu)]$$

Notice that $d(y, \mu) = 0$ only when $y = \mu$ and otherwise, $d(y, \mu) > 0$. In fact, $d(y, \mu)$ increases as μ moves away from y in either direction. This shows that $d(y, \mu)$ can be interpreted as a type of distance measure between y and μ.

GLM: Summary

EDM	$V(\mu)$	$\kappa(\theta)$	θ	ϕ	$d(y,\mu)$	S	Ω	Θ
Normal	1	$\theta^2/2$	μ	σ^2	$(y-\mu)^2$	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R}$
Binomial	$\mu(1-\mu)$	$\log(1+e^{\theta})$	$\log\frac{\mu}{1-\mu}$	$\frac{1}{m}$	$2\left\{y\log\frac{y}{\mu}+(1-y)\log\frac{1-y}{1-\mu}\right\}$	$\frac{0,1,\ldots m}{m}$	$(0,1)$	$\mathbb{R}$
Negative binomial	$\mu+\frac{\mu^2}{k}$	$-\log(1-\exp\theta)$	$\log\frac{\mu}{\mu+k}$	1	$2\left\{y\log\frac{y}{\mu}-(y+k)\log\frac{y+k}{\mu+k}\right\}$	$\mathbb{N}_0$	$\mathbb{R}^+$	$\mathbb{R}^-$
Poisson	μ	$\exp\theta$	$\log\mu$	1	$2\left\{y\log\frac{y}{\mu}-(y-\mu)\right\}$	$\mathbb{N}_0$	$\mathbb{R}^+$	$\mathbb{R}$
Gamma	μ^2	$-\log(-\theta)$	$-\frac{1}{\mu}$	ϕ	$2\left\{-\log\frac{y}{\mu}+\frac{y-\mu}{\mu}\right\}$	$\mathbb{R}^+$	$\mathbb{R}^+$	$\mathbb{R}$
Inverse Gaussian	μ^3	$-\sqrt{-2\theta}$	$-\frac{1}{2\mu^2}$	ϕ	$\frac{(y-\mu)^2}{\mu^2 y}$	$\mathbb{R}^+$	$\mathbb{R}^+$	$\mathbb{R}_0^-$
Tweedie ($\xi\le 0$ or $\xi\ge 1$)	μ^{ξ}	$\frac{\{(1-\xi)\theta\}^{(2-\xi)/(1-\xi)}}{2-\xi}$ for $\xi\neq 2$	$\frac{\mu^{1-\xi}}{1-\xi}$ for $\xi\neq 1$	ϕ	$2\left\{\frac{\max(y,0)^{2-\xi}}{(1-\xi)(2-\xi)}-\frac{y\mu^{1-\xi}}{1-\xi}+\frac{\mu^{2-\xi}}{2-\xi}\right\}$ for $\xi\neq 1,2$	$\xi<0$: $\mathbb{R}$ $1<\xi<2$: $\mathbb{R}_0^+$ $\xi>2$: $\mathbb{R}^+$	$\mathbb{R}^+$ $\mathbb{R}^+$ $\mathbb{R}^+$	$\mathbb{R}_0^+$ $\mathbb{R}^-$ $\mathbb{R}_0^-$

Binary Response: Logistic Regression

- Consider $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$, where $p_i = p_i(x_i) = \text{P}(Y_i = 1 \mid X_i = x_i)$.
- We could use a linear probability model (LPM) with the usual assumptions:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{i,p} + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

- The expected values for the linear probability model from RHS are:

$$E(Y_i | X_i = x_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{i,p}$$

- But for a Bernoulli outcome from the LHS we have:

$$E(Y_i | X_i = x_i) = p_i = \text{P}(Y_i = 1 \mid X_i = x_i)$$

Problems with Linear Regression

- Predicted values from linear regression are not constrained to the interval [0,1], even though probabilities should be.
- Residuals from the linear model are not normally distributed because they are dichotomous conditional $X = x$:

$$\varepsilon_i = \begin{cases} -p(x) \text{ if } Y_i = 0 \\ 1 - p(x) \text{ if } Y_i = 1 \end{cases}$$

- Also, the variance of ε_i is $p(x)[1 - p(x)]$, which means it is not constant.
- We need a method that deals with the above-mentioned deficiencies.

Logistic Regression: Foundation

- Assume the outcome Y is coded as 0/1. Our goal is to specify a model for

$$E(Y|x) = \Pr(Y = 1|X = x) = p(x)$$

- In linear regression we use a linear model in the covariates $X_1, \ldots, X_p$:

$$p(\boldsymbol{X}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

- The model above, however, has a range equal to $(-\infty, \infty)$. To restrict our range to (0,1), we need a function f: $(-\infty, \infty) \rightarrow (0,1)$. Ideally, the function will be simple to write, continuous, and monotonic. The *logistic function* is a prime candidate:

$$f(t) = \frac{\exp(t)}{1 + \exp(t)} = \frac{1}{1 + e^{-t}}$$

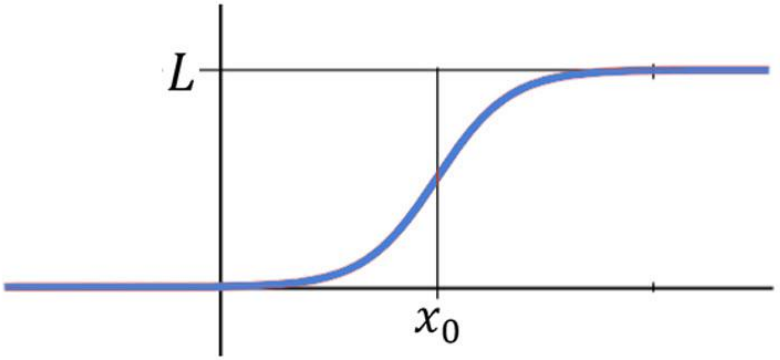
Logistic/Sigmoid Function

The sigmoid function resembles an *S*-shaped curve. It takes the real numbered input values and converts them between 0 to *L* (by shrinking from both sides, i.e., the negative values to 0 and very high positive ones to *L*). Note that it can be decreasing as well.

Logistic Function

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}}$$

x_0 = x value of midpoint
L = maximum value
k = growth rate

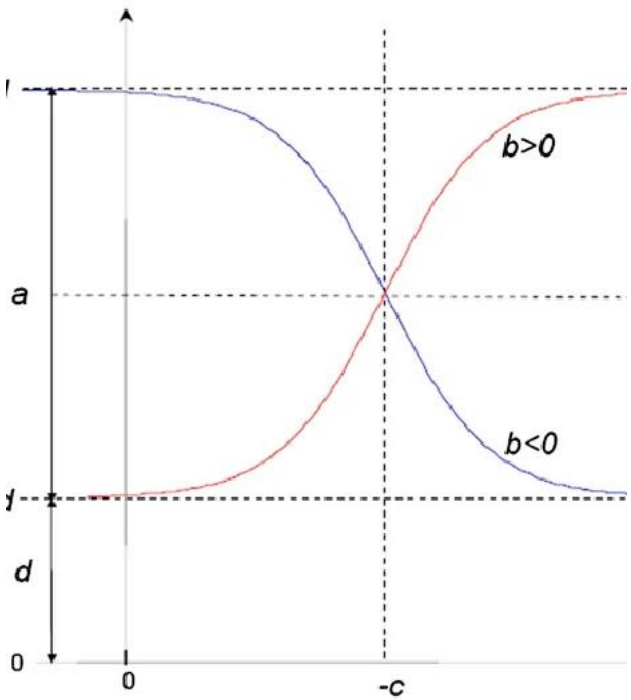


Other Response Functions

- A sigmoid function is a function having a characteristic "S"-shaped curve or sigmoid curve.

$$f(x) = \frac{a}{1 + e^{-b(x+c)}} + d$$

- The *logistic function* is a prime candidate:



- An alternative is the *probit function*:

$$\Phi(t)$$

where Φ is the cdf of the standard normal distribution. Probit has tails that die off faster, which can behave differently with rare events compared to the logistic distribution.

- Theoretically, any cumulative distribution function can serve as a response function.

Logistic Regression: Foundation

- Applying the logistic function to the linear transformation of the predictors gives three equivalent formulations.

1. probability

$$p(X) = \frac{\exp\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p\right)}{1 + \exp\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p\right)}$$

2. odds

$$\frac{p(X)}{1 - p(X)} = \exp\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p\right)$$

3. log (odds) or "logit"

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

Exercise: Show algebraically that 3 implies 1.

Example: Bank Marketing

Data in `15-bank-full.csv` come from here:
https://archive.ics.uci.edu/dataset/222/bank+marketing

- The data are related with direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, to access if the product (bank term deposit) would be ('yes') or not ('no') subscribed.
- Explanatory variable thought possibly to affect the outcome were:
- Job occupation: admin.', 'blue-collar', 'entrepreneur', 'housemaid', 'management', 'retired', 'selfemployed', 'services', 'student', 'technician', 'unemployed', 'unknown',
- Housing: has housing loan or no,
- Balance: average annual balance
- Marital status
- …

The data has 45,211 rows.

Logistic Regression Example

Use "`glm`" function to fit GLM models. "family" option determines the distribution to be used. Choices are:

- binomial(link = "logit")
- gaussian(link = "identity")
- Gamma(link = "inverse")
- inverse.gaussian(link = "1/mu^2")
- poisson(link = "log")
- quasi(link = "identity", variance = "constant")
- quasibinomial(link = "logit")
- quasipoisson(link = "log")

See R code

Logistic Regression Example: Some Results

```
> logit0 = glm(y ~ balance, data = bank, family = "binomial")
> summary(logit0)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.081e+00  1.595e-02 -130.50   <2e-16 ***
balance      3.958e-05  3.840e-06   10.31   <2e-16 ***
```

Equation: log(odds of subsrcibing) = -2.081 + 0.00003958*balance

Interpretation of the slope:

- If the balance increases by 1 euro, then the log of the odds (of subscribing) will increase by 0.00003958 units.
- Equivalent (and better!): the odds of subscribing will be multiplied by $e^{0.00003958} = 1.00004$ for each extra 1 euro of balance.
- **Final interpretation (resale)**: For each extra 1000 euros of balance the odds of obtaining the product increase by 4%

Logistic Regression Example: Dichotomous Covariate

Let's use only the housing variable

R output:

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.60687    0.01892  -84.93   <2e-16 ***
housingyes  -0.87696    0.03030  -28.95   <2e-16 ***
```

$$\log\left(\frac{\hat{p}(X_i)}{1-\hat{p}(X_i)}\right) = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

- Report the regression equation:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -1.60687 - 0.87696 * \text{Housing}$$

Logistic Regression Example: Dichotomous Covariate

Compare the equation to cross-tabs results:

No housing loan prob of subscribing = 16.7%

With housing loan prob of subscribing = 7.7%

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \texttt{-1.60687-0.87696} * \text{Housing}$$

Calculate probability, odds and log-odds for each category:

No housing loan		Housing loan
?	Probability	?
?	Odds	?
?	Log-odds	?

Logistic Regression Example: Dichotomous Covariate

- The odds can be interpreted as the relative risk of an event *A* happening versus not happening. That is,

$$odds(A) = \frac{prob(A)}{1 - prob(A)}$$

Odds = 0.2 = 1:5 means likelihood of subscribing to the service is 0.2 times the likelihood of not subscribing, within the no housing loan group. Or, you can say that the not subscribing is 5 times more likely than subscribing within the no housing group.

Exercise: Interpret the housing loan = yes group odds.

- The odds ratio is the ratio of odds for cases with different *x* values.

Odds ratio = $\frac{0.200514}{0.083423} = 2.4036$

This means that in the absence of a house loan, the odds of subscribing are 2.4 times higher, compared to the group which have a house loan.

Logistic Regression: Estimating Betas

- Just as in the simple regression, the coefficients β_0 and β_1 are unknown and must be learned using the available data. *Maximum likelihood* is usually the method for this problem.
- When $Y_i|x_i \sim Bernoulli\big(p(x_i)\big)$, then the likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p(x_i)^{y_i}[1 - p(x_i)]^{1-y_i}$$

Since

$$\log \frac{p(x_i)}{1 - p(x_i)} = \beta_0 + \beta_1 x_i$$

$$\Rightarrow \frac{p(x_i)}{1 - p(x_i)} = e^{\beta_0 + \beta_1 x_i} \Rightarrow p(x_i) = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

$$\Rightarrow 1 - p(x_i) = \frac{1}{1 + e^{\beta_0 + \beta_1 x_i}}$$

Logistic Regression: Estimating Betas

- When $Y_i|x_i \sim Bernoulli\big(p(x_i)\big)$, then the likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p(x_i)^{y_i}[1 - p(x_i)]^{1-y_i}$$

$$p(x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}}, 1 - p(x_i) = \frac{1}{1 + e^{\beta_0+\beta_1 x_i}}$$

and therefore

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} \left(\frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}}\right)^{y_i} \left(\frac{1}{1 + e^{\beta_0+\beta_1 x_i}}\right)^{1-y_i}$$

$$= \prod_{i=1}^{n} \left[\left(e^{\beta_0+\beta_1 x_i}\right)^{y_i} \frac{1}{1 + e^{\beta_0+\beta_1 x_i}}\right]$$

$$\Rightarrow l(\beta_0, \beta_1) = \sum_{i=1}^{n} y_i(\beta_0 + \beta_1 x_i) - \sum_{i=1}^{n} \log\left(1 + e^{\beta_0+\beta_1 x_i}\right)$$

Logistic Regression: Estimating Betas

- The logistic regression loglikelihood is:

$$l(\beta_0, \beta_1) = \sum_{i=1}^{n} y_i(\beta_0 + \beta_1 x_i) - \sum_{i=1}^{n} \log\left(1 + e^{\beta_0 + \beta_1 x_i}\right)$$

$$\Rightarrow \frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} = \sum_{i=1}^{n} \left(y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) = 0$$

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^{n} \left(y_i x_i - \frac{x_i e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) x_i = 0$$

- Unlike the closed-form analytical solution (i.e., the normal equations) available for linear regression, the score function for logistic regression is transcendental. That is, there is no closed-form solution so we must solve with numerical optimization methods such as Newton's method (IRLS) which is done in several iterations.
- Although we have focused on simple logistic regression, this likelihood (and score) function generalize directly to the case of *p* predictors.

Logistic Regression: Estimating Betas

- The simple logistic regression estimating equations are:

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} = \sum_{i=1}^{n} \left(y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) = 0$$

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^{n} \left(y_i x_i - \frac{x_i e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) x_i = 0$$

- In general, the gradient is:

$$\frac{\partial l(\boldsymbol{\beta})}{\partial \beta_j} = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) x_{ij}$$

- Gradient Ascent Optimization:

$$\beta_j^{new} = \beta_j^{old} + \eta \frac{\partial l\left(\boldsymbol{\beta}^{old}\right)}{\partial \beta_j^{old}}$$

where η is the magnitude of the step size that we take.

Deviance

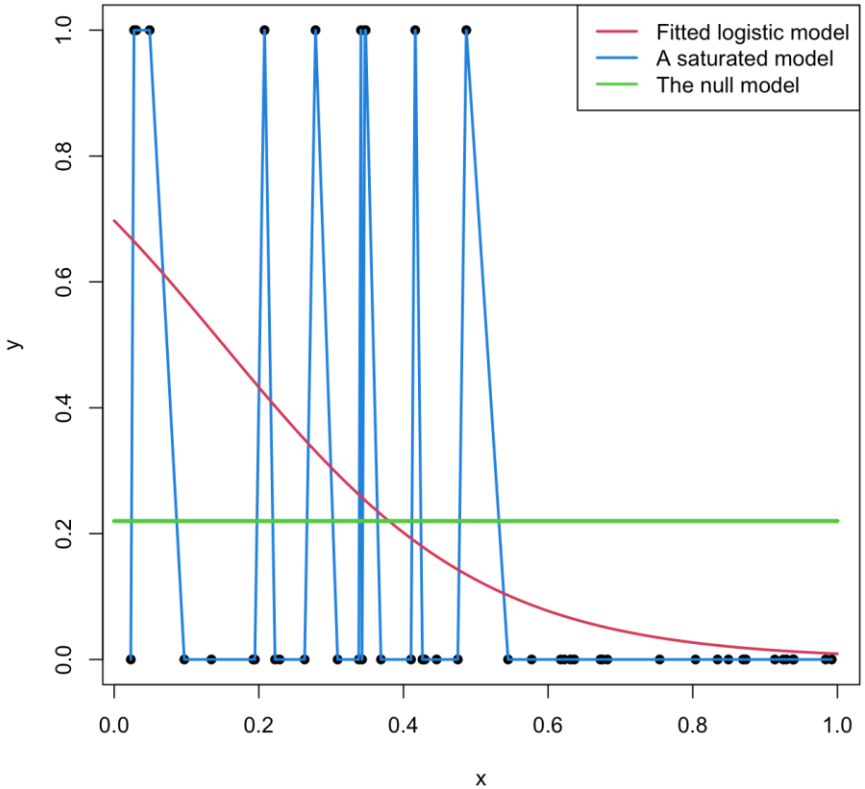
The deviance for a GLM m is defined as

$$D(\boldsymbol{y}, m) = -2\left[l(\widehat{\boldsymbol{\beta}}) - l_S\right]$$

It measures the difference of the fitted generalized linear model with respect to a perfect model for the sample.

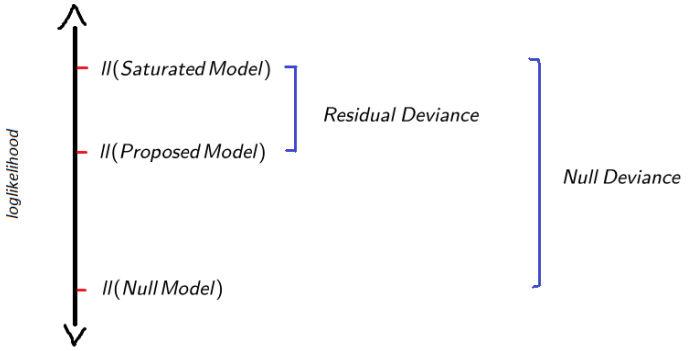
This perfect model, known as the *saturated model*, is the model that perfectly fits the data, in the sense that the fitted responses equal the observed responses.

For a linear model, deviance is the sum of squared errors (SSE) and null deviance D0 is the total sum of squares (SST).



Deviance (continued)

- It measures the difference of the fitted generalized linear model with respect to a perfect model for the sample.
- This perfect model, known as the *saturated model*, is the model that perfectly fits the data, in the sense that the fitted responses equal the observed responses.
- In the linear case, D = SSE.
- In R, the "Residual Deviance" is two times the difference in the log likelihood of the saturated (perfect) model and our model.
- In R, Null Deviance = 2[LL(Saturated Model) - LL(Null Model)]
- Note, deviance is not equal to KL distance
- But $D(\boldsymbol{y}, m_2) - D(\boldsymbol{y}, m_1) = KL(\boldsymbol{y}, m_2) - KL(\boldsymbol{y}, m_1)$
- See Hastie (1987) article in The American Statistician for more details.



Logistic Regression: Model Selection

- In multiple linear regression there were many selection criteria available: R^2, R^2-adjusted, AIC, BIC, …
- R^2 is based on the underlying assumption that you are fitting a linear model. If you aren't fitting a linear model, you shouldn't use it!
- Only AIC and BIC can be adapted to the GLM with the same formula but now using the logistic likelihood:

$$\text{AIC} = -2\log L + 2p$$

$$\text{BIC} = -2\log L + \log(n)\, p$$

- Rule is the same: smaller AIC or BIC means better model.
- **Note**: `regsubsets` function from the `leaps` package is applicable *only* for linear regression with *normal* errors. For GLM you must use `bestglm` function from the `bestglm` package.
- See R code for the healthcare example